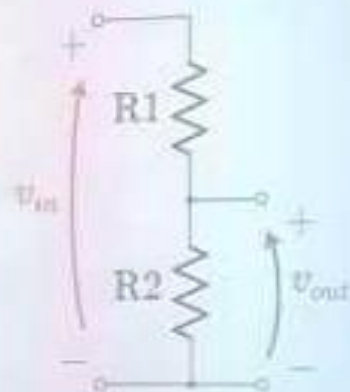


VOLTAGE DIVIDER (NO-LOAD CONDITIONS)

- A voltage divider is a passive circuit used to create a lower voltage from a higher voltage source.
- It operates on Ohm's Law and Kirchhoff's Voltage Law, dividing input voltage based on resistor values.
- The output voltage can be calculated using the Voltage Divider Rule:

$$v_{out} = v_{in} \frac{R_2}{R_1 + R_2}$$



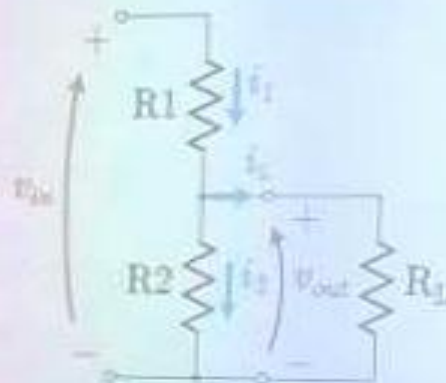
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LOADING EFFECT

When a load resistor R_L is connected to a voltage divider, the following effects occur:

- The output voltage drops, and the amount of decrease depends on the value of the load resistor R_L .
- After connecting the load, the voltage divider becomes a series-parallel circuit. As a result, the total resistance of the circuit decreases.
- Because the total resistance is reduced, the total current drawn from the source increases.



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DESIGN CRITERIA TO MINIMIZE THE LOADING EFFECT

- Resistor R_L acts as a load on the output of the voltage divider, meaning it causes a current i_L to flow. The presence of R_L means R_2 and R_L are no longer strictly in series.
- We want i_L to be small (we need it to be much smaller than i_1 or i_2), so let's make R_L fairly large (at least 10 times larger).

$$R_L = 10R_2$$

- R_2 and R_L are in parallel:

$$R_2 \parallel R_L = \frac{R_2 \cdot R_L}{R_2 + R_L}$$

$$\frac{R_2 \cdot 10R_2}{R_2 + 10R_2} = \frac{10}{11} R_2 = 0.91 R_2$$

